# Clémence Réda | Researcher @ IBENS

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## Professional Experience

### Research Positions

**Chargé de recherche* CNRS 51** | **Dr. Auguste Genovesio** — **Paris, France**
***Institut de Biologie de l'École Normale Supérieure*** *(UMR 8197)* — *02/2025–present*

**Marie Skłodowska-Curie Postdoctoral Fellow** | **Pr. Olaf Wolkenhauer** — **Rostock, Germany**
***Universität Rostock*** *(SBI Rostock)* — *05/2023–02/2025*
Development of the RECeSS project, focusing on the development of new, improved techniques for drug development based on collaborative filtering approaches.
**Skills** Collaborative Filtering · Python (Programming Language) · Applied Machine Learning

**MSC PF (research visit)** | **Pr. Hisashi Kashima & Dr. Koh Takeuchi** — **Kyoto, Japan**
***Kyoto University*** *(Machine Learning and Data Mining Research Laboratory)* — *12/2024–01/2025*
Joint project on diversity for recommendations in collaborative filtering (part of the RECeSS project).
**Skills** Collaborative Filtering · Python (Programming Language) · Applied Machine Learning

**MSC PF (secondment)** | **Dr. Jill-Jênn Vie** — **Saclay, France**
***Inria Saclay*** *(SODA team)* — *07/2023–10/2023*
Design of the JELI algorithm, a collaborative filtering approach integrating graph priors to enable explicit interpretability, and application to drug repurposing (part of the RECeSS project).
**Skills** Factorization Machines · Knowledge Graph · Interpretability

**Postdoctoral position** | **Pr. Andrée Delahaye-Duriez** — **Paris, France**
***Neurodiderot*** *(UMR 1141)* — *09/2022–03/2023*
Development and implementation of the NORDic pipeline for Boolean networks, Prefiguration of the multiomics workflow for the RHU FAME project led by Pr. Élie Azoulay.
**Skills** Systems Biology · Programming · Interdisciplinary Research · Bioinformatics

**PhD position** | **Pr. Andrée Delahaye-Duriez & Dr. Émilie Kaufmann** — **Paris, France**
***Neurodiderot*** *(UMR 1141)* & ***SCOOL*** *(UMR 9189)* — *09/2019–09/2022 (36 months)*
Combination of gene regulatory networks and sequential machine learning for drug repurposing.
**Skills** Systems Biology · Multi-Armed Bandits · Interdisciplinary Research · Bioinformatics

**Master internship** | **Pr. Andrée Delahaye-Duriez & Dr. Émilie Kaufmann** — **Paris, France**
***Neurodiderot*** *(UMR 1141)* & ***SCOOL*** *(UMR 9189)* — *03/2019–08/2019 (4 months)*
Design of a drug repurposing method through a bandit algorithm combined with the prediction of transcriptomic states by a gene regulatory network. Application to the prediction of new anti-epileptics.
**Skills** Interdisciplinary Collaboration · Interdisciplinary Research · Statistical Learning · Project Design · Bioinformatics

**Predoctoral internship** | **Dr. Bartek Wilczyński** — **Warsaw, Poland**
***Regulomics team*** *(MIM UW)* — *10/2017–07/2018 (10 months)*
Proof-of-concept on the explicit inclusion of biological interactions in gene regulatory networks and its impact on inference and simulation of transcriptomic regulation. Led to a publication in Journal of Theoretical Biology (DOI : 10.1016/j.jtbi.2019.110091).
**Skills** Network Analysis · Epigenetics · Python (Programming Language) · Systems Biology · Scientific Presentation

**Master internship** | **Dr. Nicholas Luscombe & Dr. Garth Ilsley** — **Onna-son, Japan**
***Genomics and Regulatory Systems Unit*** *(OIST)* — *02/2017–07/2017 (5 months)*
Design and implementation of a single-cell RNA sequencing clustering method taking into account intergene expression dependencies using a probabilistic model ; implementation in R Shiny of a web application for the visualisation and preliminary analysis of single-cell RNA sequencing data. Application to transcriptomic data analysis in ciona (*Ciona intestinalis*).
**Skills** Benchmarking · R Shiny · Unsupervised Learning · Data Visualization · Python (Programming Language)

**Bachelor internship** | **Dr. Macha Nikolski & Dr. Mathieu Raffinot** **Bordeaux, France**
***Centre de Bioinformatique de Bordeaux*** *(Université de Bordeaux)* *05/2016–07/2016 (2 months)*
Design and implementation of compared analyses of taxonomic trees built from metagenomic data. Application to the analysis of data from intestinal guts of children afflicted with cystic fibrosis at Hôpital Pellegrin in Bordeaux.
**Skills** Metagenomics · Phylogenetics · Supervised Learning · Unsupervised Learning · Python (Programming Language)

### Teaching & Mentoring Experiences

**Co-supervision of a Master's degree internship** **IBENS**
*Joint supervision of Raúl Durán de Alba with Dr. Olivia Lenoir (75%)* *02/2026–07/2026 (6 months)*
Analysis of biological pathway networks using artificial intelligence to better understand focal segmental glomerulosclerosis.

**Co-supervision of a Master's degree internship** **Inserm Neurodiderot**
*Joint supervision of Fabien Romano with Pr. Andrée Delahaye-Duriez (50%)* *02/2025–07/2025 (5 months)*
Blood derived multi-omics analyses to identify gene regulatory networks associated to post-traumatic stress syndrome.

**Co-supervision of a PhD** **SBI Rostock**
*Joint supervision of Orell Trautmann with Pr. Olaf Wolkenhauer (75%)* *08/2024–present*
Knowledge graphs applied to drug combinations and missing data: paper1, paper2.

**Co-supervision of a PhD** **SBI Rostock**
*Joint supervision of Rahul Bordoloi with Pr. Olaf Wolkenhauer (75%)* *09/2023–present*
Missing data for classification: paper1, paper2.

**Biostatistics, programming and bioinformatics** **Université Paris Cité**
*Doctorant Contractuel avec Mission d'Enseignement (DCME) (Teaching Assistant)* *09/2020–09/2021 (64 hours)*
**References:** Dr. Anne Badel & Pr. Olivier Taboureau

**Co-supervision of a Master's degree internship** **Inserm Neurodiderot**
*Joint supervision of Adrien Dufour with Pr. Andrée Delahaye-Duriez (25%)* *02/2020–07/2020 (6 months)*
Identification of functional families of migroglia cells from targeted single-cell RNA sequencing data of inflammatory microglia at a developmental stage: paper.

**Co-supervision of a Masters's degree project** **ENS Paris-Saclay**
*Joint supervision of Ariane Alix with Dr. Émilie Kaufmann (50%)* *11/2019–01/2020 (2 months)*
Proposal of a project on the adaptation of a published drug-target prediction method to drug repurposing using collaborative filtering in the course *Graphs in Machine Learning* taught by Dr. Michał Valko in Master Vision Apprentissage (**MVA** 2020).

* *Research-only tenured position which is equivalent to Associate professor.*

## Education

**Université Paris Cité, Inserm UMR 1141 & CNRS UMR 9189**
*PhD in Genetics* *09/2019 – 09/2022*
Doctorate Degree in Science. Title: **Combination of gene regulatory networks and sequential machine learning for drug repurposing**, supervised by Pr. Andrée Delahaye-Duriez (Inserm UMR 1141) & Dr. Émilie Kaufmann (CNRS UMR 9189).
***Viva*: 09/09/2022**.

**École Normale Supérieure† (ENS) Paris-Saclay** **(ex-École Normale Supérieure de Cachan)**
*M2 Master Vision, Apprentissage (MVA)* *09/2018 – 09/2019*
Master's degree in Machine Learning. (*summa cum laude*, Grade: 16.17/20, no ranking)

**ENS Paris-Saclay**
*M1 Master Parisien en Recherche en Informatique (MPRI)* *09/2016 – 09/2017*
Master's degree in Computer Sciences. (*summa cum laude*, Grade: 16.72/20, rank: 3/25)

**École Normale Supérieure de Cachan**
*L3 Licence informatique fondamentale ENS Cachan* *09/2015 – 09/2016*
Bachelor's degree in Computer Sciences. (*cum laude*, Grade: 14.64/20, rank: 10/26)

† *École Normale Supérieures are selective French schools for research and teaching.*

## Funding and Awards as Principal Recipient

**Accessit from the Societe Savante Francophone d'Apprentissage Machine** **SSFAM**
*PhD award (award list)* *2024*

**Marie Skłodowska-Curie Postdoctoral Fellowship 2022** **Horizon** 2020
*Postdoctoral grant* *2023–2025 (2 years)*
RECeSS project, Project ID: 101102016.

# Research

## Preprints

**Adaptive Quality-Diversity Trade-offs for Large-Scale Batch Recommendation**
C. Réda, T. Rigaux, H. Bederina, K. Takeuchi, H. Kashima, J.-J. Vie
, *Under review, HAL: 05486558*

**Handling Missing Data in Downstream Tasks With Distribution-Preserving Guarantees**
R. Bordoloi*, S. Bej, O. Wolkenhauer, C. Réda*
, *Under review, HAL: 05067568*

## Peer-Reviewed Scientific Journals

### 2026

**Embedding Learning on Multiplex Networks for Link Prediction**
O. Trautmann, O. Wolkenhauer, C. Réda.
, *Artificial Intelligence Review 59, 196, DOI: 10.1007/s10462-026-11656-w*

**An Anytime Algorithm for Good Arm Identification**
M. Jourdan, A. Delahaye-Duriez, C. Réda
, *Journal of Machine Learning Research 27(19), 1-90, DOI: 10.48550/arXiv.2310.10359*

**A systematic scoring system to optimise the testing of neurotherapeutics in models of perinatal brain injury, with an applied case study of human umbilical-cord MSC** C. Bokobza*, C. Réda*, S. Nair*, et al.
, *Journal of Neuroinflammation 23(1), 3, DOI: 10.1186/s12974-025-03593-2*

### 2025

**Multivariate functional linear discriminant analysis for partially-observed time series**
R. Bordoloi, C. Réda, O. Trautmann, S. Bej, O. Wolkenhauer
, *Machine Learning, 114, 80, DOI: 10.1007/s10994-025-06741-0*

**Comprehensive evaluation of pure and hybrid collaborative filtering in drug repurposing**
C. Réda, J.-J. Vie, O. Wolkenhauer
, *Scientific Reports, 15, 2711, DOI: 10.1038/s41598-025-85927-x*

**Joint Embedding-Classifier Learning for Interpretable Collaborative Filtering**
C. Réda, J.-J. Vie, O. Wolkenhauer
, *BMC Bioinformatics, 26, 26, DOI: 10.1186/s12859-024-06026-8*

**Neonatal inflammation impairs developmentally-associated microglia and promotes a highly reactive microglial subset**
A. Dufour*, A. Heydari-Olya*, S. Foulon*, C. Réda, et al.
, *Brain, Behavior, and Immunity, DOI: 10.1016/j.bbi.2024.09.019*

### 2024

**stanscofi and benchscofi: a new standard for drug repurposing by collaborative filtering**
C. Réda, J.-J. Vie, O. Wolkenhauer
, *Journal of Open Source Software, 9(93):5973, DOI: 10.21105/joss.05973*

### 2023

**NORDic: a Network-Oriented package for the Repurposing of Drugs**
C. Réda & A. Delahaye-Duriez
, *Journal of Open Source Software, 8(90):5532, DOI: 10.21105/joss.05532*

### 2021

**Machine learning applications in drug development**
C. Réda, É. Kaufmann & A. Delahaye-Duriez
, *Computational and Structural Biotechnology Journal, 18:241-252, DOI: 10.1016/j.csbj.2019.12.006*

### 2020

**Automated inference of gene regulatory networks using explicit regulatory modules**
C. Réda & B. Wilczyński
, *Journal of Theoretical Biology, 486:110091, DOI: 10.1016/j.jtbi.2019.110091*

## Peer-Reviewed Conference Proceedings

### 2022

**Near-optimal Collaborative Learning in Bandits**
C. Réda, S. Vakili, É. Kaufmann
, *Proceedings of the* $36^{\text{th}}$ *Conference on Advances in Neural Information Processing Systems (**NeurIPS** 2022)*
*HAL: 03825099* [*Selected as Oral*]
**Prioritization of Candidate Genes Through Boolean Networks**
C. Réda, A. Delahaye-Duriez
, *Proceedings of the* $20^{\text{th}}$ *International Conference on Computational Methods in Systems Biology (**CMSB** 2022)*
Springer:89-121 [*Best Student Paper Award*]

**2021**

**Dealing With Misspecification In Fixed-Confidence Linear Top-m Identification**
C. Réda, A. Tirinzoni & R. Degenne
, *Proceedings of the* $35^{\text{th}}$ *Conference on Neural Information Processing Systems (**NeurIPS** 2021), 34, HAL: 03409205*

**Top-$m$ identification for linear bandits**
C. Réda, É. Kaufmann & A. Delahaye-Duriez
, *Proceedings of the* $24^{\text{th}}$ *International Conference on Artificial Intelligence and Statistics (**AISTATS** 2021), 130*
*HAL: 03172145*

## Oral Communications at International Conferences

**R. Bordoloi. Multivariate functional linear discriminant analysis for partially-observed time series**
*Association for the Advancement of Artificial Intelligence (AAAI) 2026 (Singapore)* *24/01/2026*

**C. Réda. Joint Embedding-Classifier Learning for Interpretable Collaborative Filtering**
*Journées Ouvertes de Biologie, Informatique et Mathématique (JOBIM) 2025 (Bordeaux, France)* *08/07/2025*

**C. Réda. Benchmarking collaborative filtering approaches to drug repurposing**
*e:Med Meeting 2023 on Systems Medicine (Berlin, Germany)* *10/10/2023*

**C. Réda. Near-optimal Collaborative Learning in Bandits**
$35^{\text{th}}$ *International Conference on Advances in Neural Information Processing Systems (New Orleans, USA)* *07/12/2022*

**C. Réda. Prioritization of Candidate Genes Through Boolean Networks**
$20^{\text{th}}$ *International Conference on Computational Methods in Systems Biology (Bucharest, Romania)* *14/09/2022*

**C. Réda. Gene network oriented drug discovery: automated inference of Boolean networks (...)**
$13^{\text{th}}$ *Conference on Dynamical Systems Applied to Biology and Natural Sciences (held virtually)* *10/02/2022*

**C. Réda. Dealing With Misspecification In Fixed-Confidence Linear Top-m Identification**
*NeurIPS@Paris* 2021 *(Paris, France)* *08/12/2021*

**C. Réda. Automated inference of gene regulatory networks using explicit regulatory modules**
*Journées Ouvertes de Biologie, Informatique et Mathématique (JOBIM) 2020 (held virtually)* *02/07/2020*

## Poster Presentations at International Conferences

**M. Jourdan. An Anytime Algorithm for Good Arm Identification**
*Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026) (Sydney, Australia)* *06/2026*

**F. Romano. Network-centric analysis of a post-traumatic stress disorder regulatory model**
*ISMB/ECCB 2025 (Liverpool, United Kingdom)* *07/2025*

**C. Réda. JELI: an interpretable embedding-learning recommender system for drug repurposing**
*ECCB 2024 (Turku, Finland)* *09/2024*

**C. Réda. JELI: an interpretable embedding-learning recommender system for drug repurposing**
*JOBIM 2024 (Toulouse, France)* *06/2024*

**C. Réda. Towards a large-scale benchmark of collaborative filtering in drug repurposing**
*SMPGD 2024 (Paris, France)* *02/2024*

**C. Réda. Drug repurposing in breast cancer by combining bandit algorithms and Boolean networks (...)**
*ISMB/ECCB* 2023 *(Lyon France)* *07/2023*

**C. Réda. Prioritization of Candidate Genes Through Influence Maximization**
*Journées Ouvertes de Biologie, Informatique et Mathématique (JOBIM 2022, Rennes, France)* *07/2022*

**C. Réda. Dealing With Misspecification In Fixed-Confidence Linear Top-m Identification**
$35^{\text{th}}$ *International Conference on Advances in Neural Information Processing Systems (NeurIPS 2022, held virtually)* *12/2021*

**C. Réda. Top-$m$ identification for linear bandits**
$24^{\text{th}}$ *International Conference on Artificial Intelligence and Statistics (AISTATS 2021, held virtually)* *04/2021*

# Open-Source Softwares & Datasets

## Softwares

### 2024

**Joint Embedding-classifier Learning for improved Interpretability (JELI)**
C. Réda
, *Zenodo, DOI: 10.5281/zenodo.12193722, GitHub: recess-eu-project/JELI*
Python package implementing an explicitly interpretable collaborative filtering

### 2023

**BENCHmark for drug Screening with COllaborative FIltering (benchscofi)**
C. Réda
, *Zenodo, DOI: 10.5281/zenodo.8241505, GitHub: recess-eu-project/benchscofi*
Python package implementing algorithms and methods from the state-of-the-art in drug repurposing with collaborative filtering

**STANdard for drug Screening by COllaborative FIltering (stanscofi)**
C. Réda
, *Zenodo, DOI: 10.5281/zenodo.8038847, GitHub: recess-eu-project/stanscofi*
Python package for the automation of the training and validation of drug repurposing with machine learning

**Network Oriented Repurposing of Drugs (NORDic)**
C. Réda
, *Zenodo, DOI: 10.5281/zenodo.7239047, GitHub: clreda/NORDic*
Python package for the inference, analysis of Boolean networks & application to drug repurposing

## Datasets

### 2023

**PREDICT**
C. Réda
, *Zenodo, DOI: 10.5281/zenodo.7982964*
Large drug repurposing dataset with open-source generation

**TRANSCRIPT**
C. Réda
, *Zenodo, DOI: 10.5281/zenodo.7982969*
Drug repurposing dataset on transcriptomic data with open-source generation

# PhD Committees and Comités de suivi de thèse

## PhD Committees

**Alexandre Tan-Lhernould** supervised by François Fages. PhD defence on 11/10/2026.(Inria LIFEWARE, Palaiseau, France)

**Gustavo Magaña-Lopez** supervised by Loïc Paulevé. PhD defence on 12/18/2025. (Inria LaBRI, Bordeaux, France)

## Comités de suivi de thèse

**Frieda Sophia Orozco-Ruiz** supervised by Laurence Calzone. Since 10/2024 (Institut Curie, Paris, France)

# Program Chair of International Conferences

**Program Chair**: International Conference on Artificial Intelligence and Statistics (AISTATS) 2021, Neural Information Processing Systems (NeurIPS) 2023;2025-2026 (Top 30% of reviewers), International Conference on Learning Representations (ICLR) 2024-2025, International Joint Conference on Artificial Intelligence (IJCAI) 2024, International Conference on Machine Learning (ICML) 2024-2025 (Outstanding Reviewer in 2025), AAAI Conference on Artificial Intelligence (AAAI) 2025, Nature machine intelligence in 2025, Journal of Chemoinformatics (2026), Intelligent Systems for Molecular Biology (ISMB) 2026, European Conference on Computational Biology (ECCB) 2026.

# Commitment to Popularization of Sciences and Law Making

## Popularization of Sciences

**11/21/2024**: Popularization paper (in French) on drug repurposing aimed at medical practitioners: C. Réda, B. Villoutreix and A. Delahaye-Duriez. **Repositionnement de médicaments** *In* La Revue du Praticien, 21 novembre 2024, 74(9);942-6 (link)

**07/05/2019**: Popularization paper (in French) on network analysis and systems biology aimed at medical practitioners: A. Delahaye-Duriez, C. Réda and P. Gressens. **Repositionnement de médicaments** *In* Identification de cibles thérapeutiques et repositionnement de médicaments par analyses de réseaux géniques In Médecine/Sciences, 5 juillet 2019, 35:515–518 (link)

**05/2023–hiatus**: **Created and published on** RECeSS project blog: progress reports on the RECeSS project and introductory blog posts on drug repurposing and collaborative filtering.

**12/2016–09/2018**: **Published on** Tryalgo [in French] : series of blog posts on known algorithms with concrete applications, aimed at high school and college students (approx. $2,400$ unique monthly users ; two of these posts constitute the Top-$2$ most visited pages.

**10/2016**: **Published on** Binaire (blog on Computer Science affiliated with French newspaper *Le Monde*) **and** The Conversation [in French] : "A.P.B. : La vie après le bac" (conjointy written with Serge Abiteboul). Explanation of the algorithm of Gale-Shapley which has been in use in a previous version of the French national web application for high school students' applications to college

### Popularization of Law-Making

**12/2016–09/2018**: **Published on** Réfléchir.fr [in French] : series of blog posts on laws passed since 2017 in France: explanation of their content and their consequences ($534$ followers on February, $24$ $2021$).